

Bootstrap — Topic 3: Grid System (★ ★ ★ Most Important Topic)

Study paper format: notes (Roman Urdu + English technical terms) + active recall Q&A at the end.

1. Grid System kya hai?

Bootstrap Grid System **Bootstrap ka dil (heart)** hai. Almost har responsive website jo Bootstrap se banti hai ye use karti hai.

Isse ye layouts ban sakte hain:

- 2-column layouts
 - 3-column cards
 - Sidebars
 - Dashboards
 - Responsive landing pages
 - Pricing sections
 - Blog layouts
-

2. Grid kaise kaam karta hai

Bootstrap page ko **12 equal columns** mein divide karta hai.

```
-----  
|1|2|3|4|5|6|7|8|9|10|11|12|  
-----
```

Har row mein **12 available spaces** hote hain, jinhe jaise chaho divide kar sakte ho — bas total **12** hona chahiye:

6 + 6	-----6----- -----6-----
4 + 4 + 4	----4---- ----4---- ----4----
3 + 9	--3-- -----9-----
2×6	2 2 2 2 2 2

3. Grid Structure (Hierarchy)

```
html
<div class="container">
  <div class="row">
    <div class="col">
      Content
    </div>
  </div>
</div>
```

Hierarchy yaad rakho:

Container → Row → Columns

Row kabhi skip mat karo.

Correct: container → row → col Wrong: container → col ❌

4. .row

Row columns ke liye ek horizontal wrapper hai — jaise ek shelf, jispar columns rakhe jate hain.

```
html
```

```
<div class="row">
  ...
</div>
```

5. `.col`

`.col` ek **equal-width column** banata hai.

Do columns:

```
html

<div class="row">
  <div class="col">A</div>
  <div class="col">B</div>
</div>
```

Har ek ko 50% milta hai.

Teen columns:

```
html

<div class="row">
  <div class="col">1</div>
  <div class="col">2</div>
  <div class="col">3</div>
</div>
```

Har ek ko 33.33% milta hai — Bootstrap width khud calculate karta hai.

6. `.col-auto`

Jab equal width nahi chahiye, aur column sirf apne content jitni width le.

```
html

<div class="row">
  <div class="col-auto">Menu</div>
  <div class="col">Main Content</div>
</div>
```

Pehla column sirf "Menu" jitni space leta hai, doosra baqi remaining space fill karta hai.

Useful for: Buttons, Icons, Labels, Navigation.

7. Fixed Columns (`.col-1` → `.col-12`)

Equal width ki bajaye, exactly specify kar sakte ho har item konse 12 mein se kitne columns le.

`col-6` ka matlab: **12 mein se 6 columns lo.**

```
html

<div class="row">
  <div class="col-6">Left</div>
  <div class="col-6">Right</div>
</div>
```

```
html

<div class="row">
  <div class="col-4">A</div>
  <div class="col-8">B</div>
</div>
```

```
html
```

```
<div class="row">
  <div class="col-3">1</div>
  <div class="col-3">2</div>
  <div class="col-3">3</div>
  <div class="col-3">4</div>
</div>
```

$3+3+3+3 = 12$ → perfect fit.

Agar total 12 se zyada ho jaye?

```
html

<div class="row">
  <div class="col-8"></div>
  <div class="col-8"></div>
</div>
```

$8 + 8 = 16$ → Bootstrap automatically doosre column ko next line par move kar deta hai.

8. Responsive Columns

Ye Bootstrap ki asli power hai — ek hi size ki bajaye, har screen ke liye alag bata sakte ho:

- Phone → full width
- Tablet → half width
- Desktop → one-third width

Breakpoints

Class	Screen Width
col-sm	≥576px
col-md	≥768px
col-lg	≥992px
col-xl	≥1200px
col-xxl	≥1400px

```
html

<div class="row">
  <div class="col-md-6">Left</div>
  <div class="col-md-6">Right</div>
</div>
```

Small screens par: Left aur Right stacked (upar-neeche). Desktop par: Left aur Right sath sath.

```
html

<div class="col-sm-6 col-lg-3">
```

Small screen par 50%, large screen par 25%. Ye cards ke liye bohat common pattern hai.

Responsive Card Grid example:

```
html
```

```
<div class="container">
  <div class="row">
    <div class="col-sm-6 col-lg-3">Card 1</div>
    <div class="col-sm-6 col-lg-3">Card 2</div>
    <div class="col-sm-6 col-lg-3">Card 3</div>
    <div class="col-sm-6 col-lg-3">Card 4</div>
  </div>
</div>
```

- Mobile: sab cards ek ke neeche ek
- Tablet: 2 cards per row
- Desktop: 4 cards per row

9. Offsets

Offsets kisi column se pehle **empty columns** add karte hain.

```
html

<div class="row">
  <div class="col-md-6 offset-md-3">Centered</div>
</div>
```

Calculation: $\text{Offset}(3) + \text{Column}(6) + \text{empty}(3) = 12$ → content center mein aa jata hai.

Useful for: forms center karna, login pages, dialog-like layouts.

10. Column Ordering

HTML change kiye bina visual order change kar sakte ho.

```
html
```

```
<div class="row">
  <div class="col order-2">First</div>
  <div class="col order-1">Second</div>
</div>
```

HTML mein "First" pehle hai, lekin screen par "Second" pehle dikhega.

Useful classes: `order-1`, `order-2`, `order-3`, `order-first`, `order-last`

Responsive ordering: `order-md-2` — mobile par normal order, desktop par reorder.

11. Nested Grids

Ek column ke andar doosra pura grid ho sakta hai.

```
html

<div class="container">
  <div class="row">
    <div class="col-8">
      <div class="row">
        <div class="col-6">A</div>
        <div class="col-6">B</div>
      </div>
    </div>
    <div class="col-4">Sidebar</div>
  </div>
</div>
```

Nested row apne parent column ke andar ek **naya 12-column grid** start karta hai.

12. Gutters (Columns ke beech spacing)

By default Bootstrap columns ke beech spacing add karta hai — is gap ko **gutter** kehte hain.

Class	Effect
<code>g-0</code>	Koi horizontal/vertical spacing nahi
<code>g-1</code>	Small gap
<code>g-4</code>	Bara gap
<code>gx-4</code>	Sirf horizontal spacing
<code>gy-3</code>	Sirf vertical spacing

13. Real Example

html

```
<div class="container">
  <div class="row g-4">
    <div class="col-sm-6 col-lg-4">
      <div class="border p-3">Card 1</div>
    </div>
    <div class="col-sm-6 col-lg-4">
      <div class="border p-3">Card 2</div>
    </div>
    <div class="col-sm-6 col-lg-4">
      <div class="border p-3">Card 3</div>
    </div>
  </div>
</div>
```

Behavior:

- Mobile (<576px): Har card full width
 - Small screens (≥576px): 2 cards per row (`col-sm-6`)
 - Large screens (≥992px): 3 cards per row (`col-lg-4`)
-

14. Cheat Sheet

Class	Purpose
<code>.row</code>	Horizontal row banata hai
<code>.col</code>	Equal-width column
<code>.col-auto</code>	Content-based width
<code>.col-1</code> → <code>.col-12</code>	Fixed column width
<code>.col-sm-*</code>	Layout apply from $\geq 576\text{px}$
<code>.col-md-*</code>	Layout apply from $\geq 768\text{px}$
<code>.col-lg-*</code>	Layout apply from $\geq 992\text{px}$
<code>.col-xl-*</code>	Layout apply from $\geq 1200\text{px}$
<code>.col-xxl-*</code>	Layout apply from $\geq 1400\text{px}$
<code>.offset-*</code>	Column se pehle empty columns
<code>.order-*</code>	Visual order change karta hai
<code>.order-first</code>	First position par move
<code>.order-last</code>	Last position par move
<code>.g-*</code>	Horizontal + vertical gutters
<code>.gx-*</code>	Sirf horizontal gutters
<code>.gy-*</code>	Sirf vertical gutters

Practice Task

Ek responsive layout banao jahan cards **mobile par 1 per row, tablet par 2 per row, desktop par 4 per row** dikhein. Sochho konse grid classes use hongi.

Active Recall — Bootstrap Grid System

| Khud se answer karne ki koshish karo, phir neech scroll karke check karo.

Q1. Bootstrap 10-column ki bajaye 12-column grid kyun use karta hai?

Q2. Agar ek row mein total column width 12 se zyada ho jaye to kya hota hai?

Q3. `.col` aur `.col-6` mein kya farq hai?

Q4. `.col-auto` kab use karoge, `.col` ki bajaye?

Q5. `col-md-6` aur `col-lg-6` mein farq explain karo.

Q6. `col-6` ko offsets ki madad se center kaise karoge?

Q7. `order-last` aur `order-md-2` ka purpose kya hai?

Q8. Nested grid kya hota hai, aur ye kyun useful hai?

Q9. `g-4`, `gx-4`, aur `gy-4` mein kya farq hai?

Q10. Grid hierarchy kya honi chahiye, aur kya galat hota hai agar row skip kar di jaye?

Q11. Ek layout banao jahan cards mobile par 1 per row, tablet par 2 per row, aur desktop par 4 per row dikhein — konsi grid classes use hongi?

Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

A1. 12 ek highly divisible number hai — ye 1, 2, 3, 4, 6, aur 12 se evenly divide ho sakta hai, isliye layouts (2-col, 3-col, 4-col, 6-col, etc.) banana bohat flexible ho jata hai. 10 utni combinations nahi deta.

A2. Bootstrap automatically extra columns ko next line par wrap/move kar deta hai — jaise `col-8 + col-8 = 16` mein doosra column neeche chala jayega.

A3. `.col` equal-width column banata hai (Bootstrap khud width calculate karta hai — available space ke hisab se), jabke `.col-6` exactly 12 mein se 6 columns (50%) leta hai, chahe doosre columns kuch bhi hon.

A4. Jab column ko content ke hisab se width chahiye ho (equal ya fixed width nahi) — jaise Menu label, button, ya icon jiski width uske content par depend kare.

A5. `col-md-6` ka layout $\geq 768\text{px}$ se apply hota hai, jabke `col-lg-6` ka layout $\geq 992\text{px}$ se apply hota hai — dono 50% width dete hain, lekin different breakpoints par activate hote hain.

A6. `col-6` ke sath `offset-md-3` use karo: `<div class="col-md-6 offset-md-3">`.
Calculation: $3 \text{ (offset)} + 6 \text{ (col)} + 3 \text{ (empty)} = 12$, jisse content center mein aa jata hai.

A7. `order-last` column ko visual order mein sabse last position par move karta hai.
`order-md-2` responsive ordering deta hai — mobile par normal order rehta hai, desktop ($\geq 768\text{px}$) par column reorder ho jata hai.

A8. Nested grid tab hota hai jab ek column ke andar dobara `.row` aur `.col` diya jaye — ye parent column ke andar apna naya 12-column grid start karta hai. Ye complex layouts (jaise ek column ke andar aur sub-layout) banane ke liye useful hai.

A9. `g-4` columns ke beech horizontal AUR vertical dono gutters deta hai. `gx-4` sirf horizontal spacing deta hai. `gy-4` sirf vertical spacing deta hai.

A10. Hierarchy: `container → row → col`. Agar row skip karke seedha container ke andar col dala jaye, to grid system properly kaam nahi karta — row ke bina Bootstrap ka grid layout system break ho jata hai.

A11. `col-12 col-md-6 col-lg-3` — mobile par (base `col-12`) full width/1 per row, tablet (`md`) par `col-6` se 2 per row, desktop (`lg`) par `col-3` se 4 per row.

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